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A deep multiple self-supervised clustering model based on autoencoder networks

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Numerous models for deep clustering have been proposed in recent times, exhibiting remarkable performance in unsupervised learning. However, they often concentrate on the features of the data itself, seldom taking into account the structure and distribution of the data during representation learning. To address this challenge, we propose a new Deep Multiple Self-supervised Clustering model, termed DMSC, which places greater emphasis on the structural distribution of the data. The proposed model effectively integrates the advantages of autoencoder and fuzzy C-Means clustering, performing multi-level clustering evaluations throughout multiple iterations of the autoencoder network training process. It leverages a gradient-like approach for data reconstruction, enabling the autoencoder to learn features more conducive to clustering, thereby enhancing clustering performance. The experimental results show that the model performs significantly better than various common clustering algorithms on datasets of different types in multiple fields. Furthermore, to boost the efficiency of the multi-layer clustering module within our model and minimize algorithmic overhead, we integrate a distance-based Two-stage fuzzy C-Means clustering method. This approach introduces an efficient, adaptable, and rational technique for initializing cluster centers and membership matrices for fuzzy C-Means clustering, achieving convergence of the loss function in a shorter time frame. Compared to the performance of traditional fuzzy C-Means clustering algorithms on various public datasets, our proposed method significantly reduces computation time and noticeably improves iteration efficiency.

Keywords Deep clustering, Autoencoder, Fuzzy C-means clustering

Clustering analysis is one of the core tasks in data mining and machine learning, aiming to partition an unlabeled dataset into several groups such that objects within the same group are as similar as possible, while objects between different groups are significantly distinct. In recent years, clustering techniques have been widely applied in various fields, such as in the medical field where related technologies are used for brain MRI image segmentation[1,2], and so on. In the context of internet-driven and big data-driven environments, a critical challenge lies in how to efficiently uncover latent information from unlabeled datasets or multi-image samples and partition them into N classes. With the advancement of computer science and artificial intelligence, an increasing number of scholars have employed mathematical principles and intelligent algorithms to achieve clustering functionality[3–5].

Traditional clustering methods mainly include hierarchical[6,7], partition-based[8,9] and density-based machine learning algorithms[10]. The earliest clustering concept emerged in the form of hierarchical clustering techniques in the 1960s[11]. The advent of hierarchical clustering also facilitated the advancement of clustering approaches. In the following years, numerous unsupervised clustering algorithms were continually proposed, such as KMEANS (K-means clustering)[12], SLINK (Single-Link Hierarchical Clustering)[13], OPTICS (Ordering Points To Identify the Clustering Structure)[14], and so forth.

Clustering tasks typically encompass four iterative steps: feature selection/extraction, clustering algorithm design/selection, cluster validation, and result interpretation. Many scholars will introduce innovation at these stages, such as the proposed algorithm AGAD (Automatic Genetic Algorithm in clustering for Discrete elements)[15], which uses a new clustering similarity index to determine the appropriate number of clusters. Some scholars have also proposed clustering algorithms that determine the number of clusters during the iterative

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process to effectively avoid getting stuck in local minima problems[16]. The fuzzy clustering algorithm based on fuzzy theory proposed by Bezdek and Dunn is also a milestone in the development history of clustering theory[17]. In recent years, there have been many innovative articles on fuzzy clustering algorithms, such as the SSFCM-FWCW (Semi-supervised fuzzy C-means method based on feature-weight and cluster-weight learning)[18], which effectively improves clustering performance by utilizing label information and adaptive weighting techniques. However, this method belongs to a semi supervised clustering method and relies on certain prior knowledge. Some scholars have also proposed a multi view clustering algorithm based on intuitionistic fuzzy c-means, which solves key issues such as noise sensitivity, outlier influence, and the importance of views, features, and samples[19]. For datasets where different features contribute differently to clustering results, using feature weighted fuzzy clustering algorithms can also significantly improve performance[20,21]. Some scholars have also proposed an improved fuzzy clustering analysis algorithm, which distinguishes geometric structures with overlapping element clusters for processing overlapping elements[22].

However, traditional clustering methods often rely heavily on pre-defined distance metrics or similarity standards for iteration, which often makes it difficult to capture the complex internal structure of the data and may not perform as expected.In recent years, due to the high complexity of real-world datasets, conventional methods often fail to yield satisfactory results. Deep neural networks, with their formidable capacity for nonlinear fitting and feature extraction, have increasingly been harnessed by researchers in the clustering domain[23–25], leading to the proposition of numerous novel clustering methodologies, such as DCN (Deep Clustering Network) and ODC (One-shot Deep Clustering)[26–28]. These approaches integrate deep embedding frameworks or employ deep neural network models. We refer to this class of innovative clustering algorithms as deep clustering.

Autoencoders, standing as one of the most quintessential neural network models, are widely employed in deep clustering due to their robust capability for unsupervised feature learning[29–32]. In this vein, scholars have proposed a seminal model known as DEC (Deep Embedding for Clustering)[33], which is grounded in the use of autoencoders, this approach realizes deep clustering by iteratively optimizing a clustering objective centered around reducing the Kullback–Leibler divergence. However, it tends to exhibit suboptimal performance when clusters are closely related or overlap significantly. Following this, researchers have proposed numerous enhanced models based on the foundation of autoencoders, aiming to address such limitations, For example, models like DCEC (Deep Convolutional Embedded Clustering)[34] and DEPICT (Deep Embedded Regularized Clustering)[35]have been put forward. DEPICT, in particular, stacks multiple layers of convolutional autoencoders and incorporates the reconstruction function as a regularization term, combining it with the clustering loss for joint training of the network. EDCWRN (Efficient Deep Clustering with the Weight of Representations and the help of Neighbors)[36], This algorithm improves clustering performance through feature weighting and neighbor information during autoencoder training.Apart from these, there are deep clustering models based on sparse autoencoders, such as DEN (Deep embedding network for clustering)[37]and DSSEC (Deep Stacked Sparse Embedded Clustering)[38]. Sparse autoencoders, compared to traditional ones, extract sparser features which augment data distinguish ability, indirectly enhancing clustering efficacy. Nonetheless, the tuning of hyperparameters governing the sparsity in sparse autoencoders lacks a universal guideline, rendering the network training process more intricate.

There are also deep clustering methods based on Variational Autoencoders[39], which serve as both generative and clustering models. Although they can leverage the power of generative models to capture the posterior probability distribution of data samples belonging to different clusters, thereby enhancing clustering performance, the integration of such generative networks comes with substantial computational demands. The use of VAEs and similar generative architectures significantly increases the need for computational resources due to the complexity involved in estimating variational lower bounds, implementing stochastic gradient descent on large datasets, and accommodating the increased capacity of these models.

While numerous models for deep clustering have already been proposed, they tend to primarily focus on the features of the data itself, rarely considering the structure and distribution of the data during representation learning[40]. Taking the autoencoder as an example, its network architecture is highly complex, consisting of multiple layers, each of which captures different latent information. There exists a variety of structural information between data points. Therefore, how to consider the data distribution when training deep clustering networks to make the output of the autoencoder more conducive to clustering, how to pull the marginal samples toward the correct classes, and achieve rapid convergence are very worthy of study. Based on this, we propose a deep multilayer clustering self-supervised model. Accordingly, we propose a self-supervised deep multi-layer clustering model.

The main contributions of this paper are as follows:

(1) We propose a new Deep Multiple Self-Supervised Clustering Model (DMSC), which places greater emphasis on the structural distribution of the data. This model effectively integrates the strengths of autoencoders and fuzzy C-Means clustering, conducting multi-level clustering evaluations over multiple iterations of the autoencoder network training. By employing a gradient-like method for data reconstruction, it enhances the focus on the structure of data distribution, thereby enabling the autoencoder to learn features more conducive to clustering, and enhancing clustering performance. The experiment shows the results demonstrate that the model significantly outperforms traditional clustering algorithms.
(2) To enhance the efficiency of the multi-level clustering module within the model and reduce algorithmic overhead, we incorporate a Two-stage clustering method based on distance. This approach employs a succinct, adaptable, and rational methodology for determining initial cluster centers for fuzzy C-Means clustering and for inputting membership matrices. It achieves convergence of the loss function in a shorter time frame compared to the original fuzzy C-Means clustering algorithm, demonstrating a substantial improvement in iteration efficiency. Extensive experiments across multiple datasets confirm a significant reduction

in both the number of iterations and runtime,resulting in an enhancement of iteration efficiency ranging from 15.7% to 63.6% compared to the original.

The structure of this paper is organized as follows. In section "Problem description", we provide background on the clustering problem addressed and define relevant variables. Section "The proposed model" introduces the fundamental techniques employed and presents our proposed model. In section "Experiment", we outline the experimental setup, detail the experimentation process, and present and interpret the results. This section also encompasses supplementary experiments related to determining initial cluster inputs and membership matrices as mentioned above,as well as sensitivity analysis of key parameters. Lastly, in the concluding section, we summarize our findings and suggest avenues for future improvements to the proposed method.

Problem description

Most clustering methods based on autoencoders rely on representation learning via a reconstruction loss function[41], with the joint optimization process described as follows:

$$L = L_1 + \gamma L_2 \tag{1}$$

where L_1 represents the reconstruction loss of the neural network, L_2 denotes the clustering loss, and γ is a hyperparameter used to balance the relationship between the two loss functions, as illustrated in Fig. 1.

However, such joint training does not facilitate optimal synergy, the implicit competition between the two aspects slows down the convergence of the optimization process. On one hand, the clustering loss aims to minimize the disparity between different clusters while preserving similarity within clusters. On the other hand, the reconstruction loss of the autoencoder solely seeks to minimize the discrepancy between all reconstructed samples and their original counterparts. Moreover, the features generated during the training of the autoencoder are not guaranteed to be conducive to clustering methodologies.

Based on these considerations, we propose a novel Deep Multiple Self-Supervised Clustering Model (DMSC), which integrates the clustering process during the training of the autoencoder network, using clustering information to reconstruct data. This reconstructed data is then fed back into the training of the autoencoder in a reverse manner. To enhance the practicality of the model, the clustering method employed is a soft clustering technique, specifically the Fuzzy C-Means algorithm. Additionally, given the substantial computational costs associated with deep clustering, our model introduces a distance-based method for determining initial cluster centers and initial membership matrices during clustering. The evaluation metrics primarily used in this study include Accuracy, Rand Index, Precision, F1-score, Normalized Mutual Information, and Recall. Table 1 provides descriptions of the symbols used in the model.

The proposed model

Overview

In this section, we present the flowchart of the proposed Deep Multiple Self-Supervised Clustering Model, as shown in Fig. 2. Our approach initially preprocesses the raw input data and feeds it into an encoder, mapping the encoded data to the hidden layer where it is input into our proposed Two-stage fuzzy C-Means clustering model. This process determines the initial membership matrix and cluster centers for fuzzy clustering. The encoded data is then clustered and undergoes unsupervised evaluation using the silhouette coefficient. If criteria are met, data distribution is adjusted to bring data points closer to their respective cluster centers. The reconstructed data is passed through the decoder. This iterative training of the autoencoder continues until optimal features

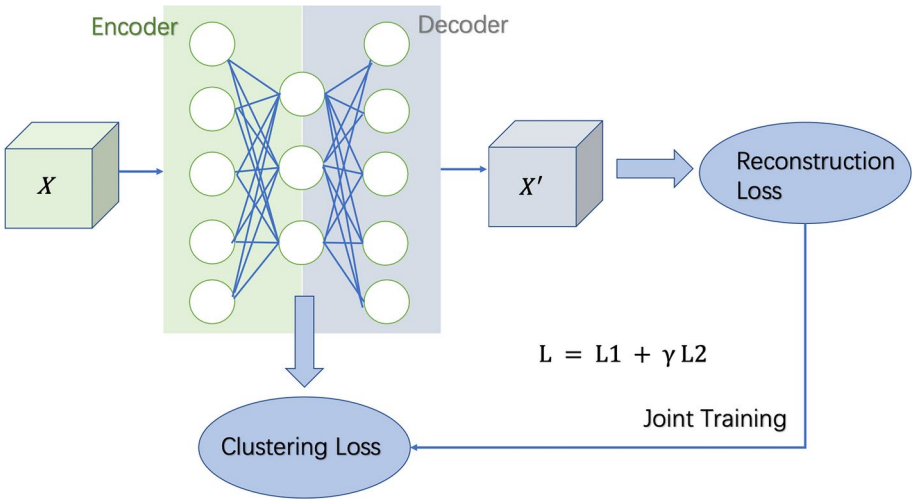

Fig. 1. Joint optimization process.

Notations	Description
X	Original dataset
M_{ij}	Distance matrix
$G_{\theta 1}$	Hidden layer mapping
U	Initial Membership matrix
i	Number of Hidden neurons
L	Loss function
Z	Encoded data
μ_{ij}	Membership weights
$C_i^{(0)}$	The i-th initial cluster center
λ	Learning rate
$H_{\theta 2}$	Decoding layer mapping
γ	Adjust parameters
Sil	Silhouette coefficient
ρ	Reconstruction threshold
C_j	Cluster center
Z_i'	Reconstructed data

Table 1. Nomenclature.

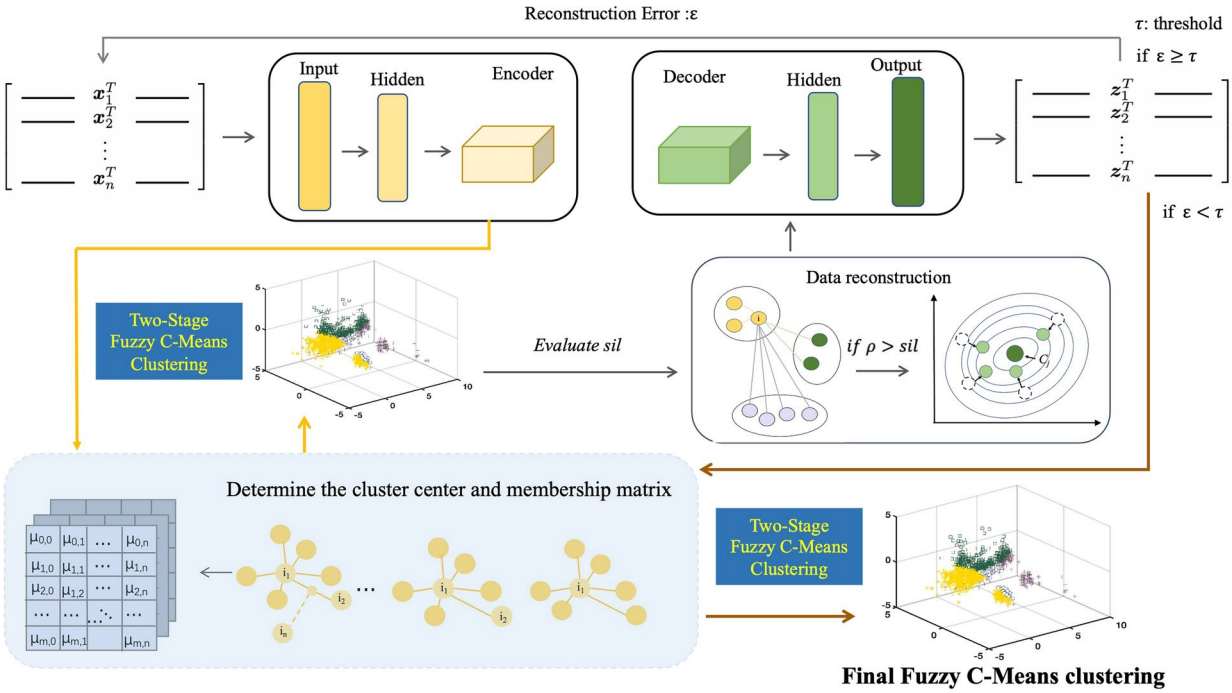

Fig. 2. Deep multiple self-supervised clustering model structure.

for clustering are outputted. Finally, the model utilizes the aforementioned Two-stage fuzzy C-Means clustering approach to yield the final results.

Baseline models overview

(1) Autoencoder. An autoencoder is an unsupervised feature learning method primarily used for data dimensionality reduction and feature extraction[42]. It utilizes the data X itself as a supervisory signal to guide the training of the neural network. After training, it attempts to replicate the input at the output, aiming for the neural network to learn the mapping $F_\theta : X \to X'$.The autoencoder splits the network F_θ into two parts. The first sub-network attempts to learn the mapping: $G_{\theta 1} : X \to Z$, while the second sub-network attempts to learn the mapping $H_{\theta 2} : Z \to X'$. The mapping $G_{\theta 1}$ is regarded as a data encoding process.The high-dimensional input X is encoded into a low-dimensional latent variable Z, referred to as the encoder network. The mapping $H_{\theta 2}$ is regarded as a data decoding process, where the encoded input Z is decoded back into the

high-dimensional X', referred to as the decoder network. Together, the encoder and decoder accomplish the encoding and decoding of the input data X. If deep neural networks are used to parameterize the functions $G_{\theta 1}$ and $H_{\theta 2}$, the model is called a deep autoencoder. To ensure that the decoded output data of the autoencoder is as close as possible to the input data, the autoencoder sets an optimization objective:

$$MinJ(G_{\theta 1}, H_{\theta 2}, X, X^{'}) = dist(X, X^{'}), X^{'} = H_{\theta 2}(G_{\theta 1}(X)) \tag{2}$$

Let dist represent the distance measure between X and X', referred to as the reconstruction error function. A common distance measure is the Euclidean distance, specifically:

$$MinJ(G_{\theta 1}, H_{\theta 2}, X, X^{'}) = 1/2||X - H_{\theta 2}(G_{\theta 1}(X))||^2 \tag{3}$$

(2) Fuzzy C-means clustering. Fuzzy C-means clustering (FCM) is a clustering algorithm based on fuzzy set theory proposed by Bezdek et al.[43]. The objective of FCM is to automatically partition a set of samples into k clusters, while also calculating the degree of membership of each sample to its respective cluster.The standard FCM clustering algorithm determines the cluster center matrix $V$and the membership matrix U by minimizing the objective function expressed as Eq. (4), the initial cluster center is usually initialized randomly.

$$\min_{U,V} \sum_{i=1}^{n} \sum_{j=1}^{k} u_{ij}^{r} \|x_i - v_j\|_2^2 \tag{4}$$

In the equation, u_{ij} represents the membership degree of the i-th sample to the j-th cluster. The membership degree is non-negative, and the sum of the membership degrees for each sample equals 1. The matrices U and V are updated through iterative optimization using Eqs. (5) and (6), until the objective function converges.

$$u_{ij} = \frac{1}{\sum_{g=1}^{k} \left(\frac{\|x_i - v_j\|_2^2}{\|x_i - v_g\|_2^2} \right)^{\frac{1}{r-1}}} \tag{5}$$

$$v_j = \frac{\sum_{i=1}^{n} u_{ij}^{r} x_i}{\sum_{i=1}^{n} u_{ij}^{r}} \tag{6}$$

Deep multiple self-supervised clustering model based on autoencoder networks

(1) Defining the autoencoder model. First, the normalized raw data is input into the autoencoder network. The encoder is defined as a sequence of linear layers and nonlinear activation functions, specifically ReLU. In detail, the dimensionality of the raw input data is reduced to an intermediate dimension M while applying the ReLU activation function. The raw data is encoded at this intermediate dimension, providing an abstract representation that captures important features and structures of the original data, aiding in data compression. However, this intermediate representation may still contain some less important information or noise. Therefore, we further compress the data to the input dimension N. Additionally, to capture the essential features of the data, there is no explicit activation function in this layer, allowing better retention of the main structural information of the data.For the decoder, it also includes two layers. The decoder maps the low-dimensional representation generated by the encoder back to the original input space to reconstruct the original data. Both layers correspond to the structure of the encoder, and finally, the Sigmoid activation function is used to ensure that the output data is within the range $[0, 1]$.

(2) Specific steps of the model. The preprocessed raw data is input into the encoder. Assume the input raw data is $X = [x_1, x_2, \ldots, x_n]$. The raw data is mapped to the hidden layer through the mapping $G_{\theta 1}$. The number of neurons in the hidden layer is i($i < n$), thus achieving compressed encoding of the raw data. The encoding process is as follows Eq. (7) where the encoded data is $Z = [z_1, z_2, \ldots, z_i]$.

$$Z = G_{\theta 1}(X) = (W_1 X + B_1) \tag{7}$$

To make the trained autoencoder generate features more suitable for clustering and thereby improve clustering performance, we introduce a clustering process during the training of the encoder. Specifically, we perform an initial Fuzzy C-Means (FCM) clustering on the encoded data. To reduce the computational cost of deep clustering algorithms and enhance clustering efficiency, we propose a method to improve the traditional FCM's initialization of cluster centers and membership matrices. This method aims to improve the iterative efficiency of clustering algorithms in the model. The steps for improving the determination of initial cluster centers and membership matrices are as follows:

① In the encoded data Z, randomly select an initial cluster center $c_1^{(0)}$ and compute its Euclidean distance to all samples in Z. Specifically, $d_{i1} = \|c_1^{(0)} - z_i\|^2$ for each sample z_i in Z. Choose the point z_p that has the maximum distance from $c_1^{(0)}$ as the second initial cluster center $c_2^{(0)}$, such that $\max[d_{i1}(z_i, c_1^{(0)})] = d_{p1}(z_p, c_1^{(0)})$.

② Find the midpoint α_1 of $c_1^{(0)}$ and $c_2^{(0)}$, calculate the Euclidean distance between α_1 and all samples in Z, and select the point z_q in Z that is farthest from α_1 as the third initial cluster center $c_3^{(0)}$, where $\max[d_{i1}(z_i, \alpha_1)] = d_{q1}(z_q, \alpha_1)$.

③ Similarly, compute the midpoint α of the current cluster centers. Calculate the Euclidean distance between α and all samples in Z, and select the sample point corresponding to the maximum d_{ij} as the new cluster center $c_j^{(0)}$.

④ Repeat step ③ until m cluster centers are found.

⑤ Calculate the Euclidean distance between the sample points and all determined initial cluster centers. This is given by $d_{ij}(z_i, c_j^{(0)}) = \|c_j^{(0)} - z_i\|^2$, resulting in the initial distance matrix $M_{(ij)}$.

⑥ Perform exponential operations on the distance matrix $M_{(ij)}$. The negative exponential function is relatively stable over the range from 0 to positive infinity, with function values between $[0, 1]$. Compute $M_{ij}^{'} = e^{-M_{ij}}$, where $M_{ij}^{'}$ is the negative exponential matrix corresponding to the initial distance matrix. The farther the sample point z_i is from the initial cluster center $c_j^{(0)}$, the smaller the corresponding value in $M_{ij}^{'}$ will be.

⑦ Calculate the initial membership matrix U. To ensure that the sum of the membership degrees of sample point z_i to the initial cluster centers $c_j^{(0)}$ equals 1, i.e., $\sum_{j=1}^{n} \mu_{ij} = 1$, μ_{ij} is shown in Eq. (8).

$$\mu_{ij} = \frac{M_{ij}^{'}}{\sum_{j=1}^{n} M_{ij}^{'}} \tag{8}$$

Next, determine the objective function for clustering. According to the concept of Fuzzy C-Means clustering, the clustering process is transformed into an optimization process for the objective function. The modeling formula is as follows:

$$\min(L) = \sum_{i=1}^{n} \sum_{j=1}^{m} \mu_{ij}^{a} \|z_i' - c_j\|^2 \tag{9}$$

Based on the objective function and constraints, using the method of Lagrange multipliers, we can derive the update formulas for the cluster centers c_j and the weights μ_{ij} by taking partial derivatives of the objective function with respect to μ_{ij} and c_j. The update formulas are given in Eqs. (11) and (12). Update μ_{ij} and c_j iteratively until the difference between successive calculations of μ_{ij} is very small or a preset number of iterations is reached. At this point, the objective function converges, and the clustering results are output.

$$L(\mu, c, \lambda) = \sum_{i=1}^{n} \sum_{j=1}^{m} \mu_{ij}^{a} \parallel z_i - c_j \parallel^2 + \sum_{i=1}^{n} \sum_{j=1}^{m} (\lambda_i \mu_{ij} - \lambda_i) \tag{10}$$

$$c_j = \frac{\sum_{i=1}^{n} \mu_{ij}^{a} z_i}{\sum_{i=1}^{n} \mu_{ij}^{a}} \tag{11}$$

$$\mu_{ij} = \frac{1}{\sum_{k=1}^{m} \left(\frac{\|z_i - c_j\|}{\|z_i - c_k\|}\right)^{\frac{1}{2(a-1)}}} \tag{12}$$

During the training of the autoencoder, to generate features more suitable for clustering, the encoded data is first clustered using the improved Fuzzy C-Means clustering method proposed in this paper. Next, the effectiveness of the clustering results is evaluated. Since the proposed model is a self-supervised model, the evaluation method used here is the silhouette coefficient method. This method entirely relies on assessing the density within clusters and the dispersion between clusters, as shown in Eq. (13). The clustering effect is evaluated by measuring the similarity of a sample with its own cluster and other clusters. α_{ij} is the average distance between the i-th sample z_i of the encoded data and other samples in the same cluster, while $\beta_{ij'}$ is the average distance between the i-th sample of the encoded data and samples in other clusters.

$$Sil = \begin{cases} 1 - \frac{\alpha_{ij}}{\beta_{ij'}}, & \text{if} \quad \alpha_{ij} < \beta_{ij'} \\ 0, & \text{if} \quad \alpha_{ij} = \beta_{ij'} \\ \frac{\beta_{ij'}}{\alpha_{ij}} - 1, & \text{if} \quad \alpha_{ij} > \beta_{ij'} \end{cases} \tag{13}$$

The silhouette coefficient ranges from $(-1, 1)$. When it approaches 1, it indicates that the sample is highly similar to others in its own cluster and dissimilar to samples in other clusters. During the training of the autoencoder, we will input a predefined threshold ρ. Once the evaluated silhouette coefficient reaches this threshold, we will reconstruct the encoded data Z. In the training process, we will use a gradient descent-like approach to fine-tune the encoded representations in each iteration, making them better approximate the cluster centroids and conform to the cluster characteristics. This optimization enhances the representation of the autoencoder, making it better reflect the clustering structure of the data. The reconstruction formula is shown in Eq. (14). Here, z_i is the encoded representation obtained from the original data through the encoder, c_j is the cluster center, the reconstructed data is z_i', and γ is the adjustment parameter $(\gamma > 0)$, which controls the adjustment magnitude. First, calculate the deviation of z_i from c_j, and then reduce the deviation accordingly.

$$z_i^{'} = z + \gamma(c_j - z) \tag{14}$$

Using this gradient-like method, we minimize the difference between the encoded representations and the cluster centroids through multiple iterations, ensuring that the encoded features better reflect the clustering structure of the data.

Next, the adjusted data z_i' is used as the new encoded data and mapped to the decoding layer for decoding, as shown in Eq. (15). The loss function and backpropagation algorithm are then used to train the autoencoder network based on deep clustering, Set the output threshold to τ and the reconstruction error to ϵ.

$$X^{'} = H_{\theta 2}(Z^{'}) = (W_2 Z^{'} + B_2) \tag{15}$$

Apply the trained deep multi-supervised clustering model to the original data to obtain the feature representations of the encoding layer and achieve dimensionality reduction. The generated feature representations also emphasize the distribution structure of the data, helping to improve the performance and generalization ability of the deep clustering model. Finally, use the proposed method of improving the initial membership matrix and initial cluster centers to perform Fuzzy C-Means clustering on the generated data. The iterative steps are the same as those from ① to ⑦. The final result of deep clustering is then output.

Experiment

Experimental setup

In this section, we will describe the dataset, comparison methods, parameter settings, and evaluation metrics to validate the effectiveness of the proposed DMSC for clustering tasks.

The autoencoder network used in the model consists of two linear layers with ReLU nonlinear activation functions embedded in each layer: $ADim_{128}^{e} \rightarrow ADim_3 \rightarrow ADim_{128}^{d}$, where Dim_n^k represents the number of neurons, indicating the dimensions of the projections; all data dimensions are $d \rightarrow 128 \rightarrow 3 \rightarrow 128$, where d is the dimensionality of the input original features. The decoder mirrors the encoder.

Training is set for 200 epochs. The clustering method embedded in our proposed model is the improved Fuzzy C-means clustering method we introduced. The experimental results of this enhanced clustering method will also be presented in section "Convergence analysis". The fuzzy coefficient of fuzzy C-means clustering is set to $\alpha = 2$, previous literature has mentioned that the optimal choice for α is within the range of [1.5, 2.5], and α is taken as the average and midpoint of this range "2" has been proven to be effective and widely used[44]. Silhouette coefficient evaluation is performed every 10 epochs to assess clustering results. The learning rate γ is set to 0.001, and the number of iterations is set to 50. Data adjustment and reconstruction are conducted when the predefined threshold is reached.

The datasets used include 9 UCI public datasets, 1 biomedical brain tumor imaging dataset, 1 movie review text dataset, and a portion of the MNIST dataset. In order to reduce computational costs, some MNIST datasets use random sampling to obtain subsets. The brain tumor dataset uses ResNet50 to extract information from image features at various levels. The introduction of each dataset is shown in Table 2. Readers can access all experimental data, please refer to the Data availability.

Comparison with clustering algorithms such as KMEANS and density-based clustering is conducted to validate the effectiveness of the proposed model.

Several well-known evaluation metrics are selected to compare the performance of our proposed model with other common clustering methods, including Accuracy, Rand Score, Precision, F1 Score, Normalized Mutual Information, and Recall Rate. In this article, ACC, RI, PRE, F1, NMI, and REC are used to denote these six evaluation metrics, and their calculation formulas are as follows:

Dataset domain	Dataset name	Number of sample	Number of attributes	Describe
Botany	Iris	150	4	Iris flower characteristics records
	Seeds	210	7	Wheat kernel characteristics records
Medicine	Diabetes	768	8	Individual records of diabetes
	Liver	345	6	Male liver disease patient records
	Sym	350	2	Individual symptom data recording
	Brain Tumor	229	2	Brain images of normals and tumor patients
Chemistry	Glass	214	6	Glass oxide content records
	Wine	178	13	Wine chemical composition analysis
Sociology	Movie Reviews	50,000	2	Movie review text data
	Cmc	1473	8	Contraceptive methods differ by population
Other	MNIST	70000	784	Handwritten digit images
	Wave	5000	21	Waveform characteristics records

Table 2. Clustering results and artificial mark contrast.

(1) Accuracy. $ACC = \frac{\sum_{i=1}^{M} N_i^j}{N}$, where N_i^j represents the count of correctly matched samples in the clustering result for the i-th object.

(2) Rand Index. $RI = \frac{(a+b)}{C_2^{n_i}}$, where a denotes the count of element pairs that are in the same cluster in both the clustering result and the actual class, b denotes the count of element pairs that are in different clusters in both the clustering result and the actual class, and $C_2^{n_i}$ is the total number of possible element pairs in the dataset.

(3) Precision. $PRE = \frac{\sum_{i=1}^{M_1} N_i^j}{\sum_{i=1}^{M_1} N_i^j + \sum_{i=1}^{M_2} F_i^j}$, where N represents the count of correctly clustered samples, and F represents the count of incorrectly predicted samples.

(4) F1 score. $F1 = \frac{2\sum_{i=1}^{M_1} N_i^j}{2\sum_{i=1}^{M_1} N_i^j + \sum_{i=1}^{M_2} F_i^j + \sum_{i=1}^{M_3} P_i^j}$, where N represents the count of correctly clustered samples, F represents false negatives, which refers to the number of instances that should have been predicted as positive but were incorrectly predicted as negative, and P represents false positives, which refers to the number of instances that should have been predicted as negative but were incorrectly predicted as positive.

(5) Normalized Mutual Information. $NMI = \frac{2I(X;Y)}{(H(X)+H(Y))}$, where $I(X;Y)$ represents the mutual information between the true class labels Y and the cluster assignments X, and H represents the entropy. $H(X) = -\sum_{i=1}^{|x|} P(i) log P(i)$.

(6) Recall Rate. $Rec = \frac{\sum_{i=1}^{M_1} N_i^j}{\sum_{i=1}^{M_1} N_i^j + \sum_{i=1}^{M_3} P_i^j}$, where N represents the count of correctly clustered samples, and P represents the count of false negatives that is incorrectly predicting positive when it should be negative.

All experiments were conducted using Matlab 2021a and Python 3.10 on a Windows PC equipped with an Intel 2.8 GHz CPU and 64GB RAM.

Experimental results and analysis

The proposed method was applied to the aforementioned datasets. The encoder part consists of two fully connected layers with a ReLU activation function. The first linear transformation outputs to a hidden layer set at 216 dimensions. The encoding layer dimensions are set to lower dimensions depending on the dataset. Parameters for the decoding layer are identical to those of the encoding layer. A common learning rate of 0.001 was used. During the data reconstruction phase, the silhouette coefficient threshold was set to $\rho = 0.5$, with 50 iterations. Experimental results are shown in Fig. 3.

To further validate the effectiveness of the proposed model, comparative experiments were conducted between the proposed method and six other methods: KMEANS, FCM, DBSCAN, SPECTRAL, OPTICS, as well as a step-by-step deep clustering approach that uses autoencoders for feature extraction followed by fuzzy C-means clustering (AUT-FCM). The experimental results are shown in Table 3, indicating that our method performs well on most datasets. Compared with KMEANS, the ACC of the eleven datasets has improved by 2% to 38%, the PRE range is 2% to 27%, and the highest improvement in NMI is 90%.

Compared to SPECTRAL clustering, although DMSC performs better than SPECTRAL clustering on most datasets, SPECTRAL clustering may perform better on certain datasets, especially those with simple cluster shapes and less noise, due to its graph based nature and efficient similarity measurement.

However, compared with FCM and methods that only use autoencoder dimensionality reduction for clustering, the proposed model exhibits stronger clustering performance. Compared with FCM, the ACC improvement

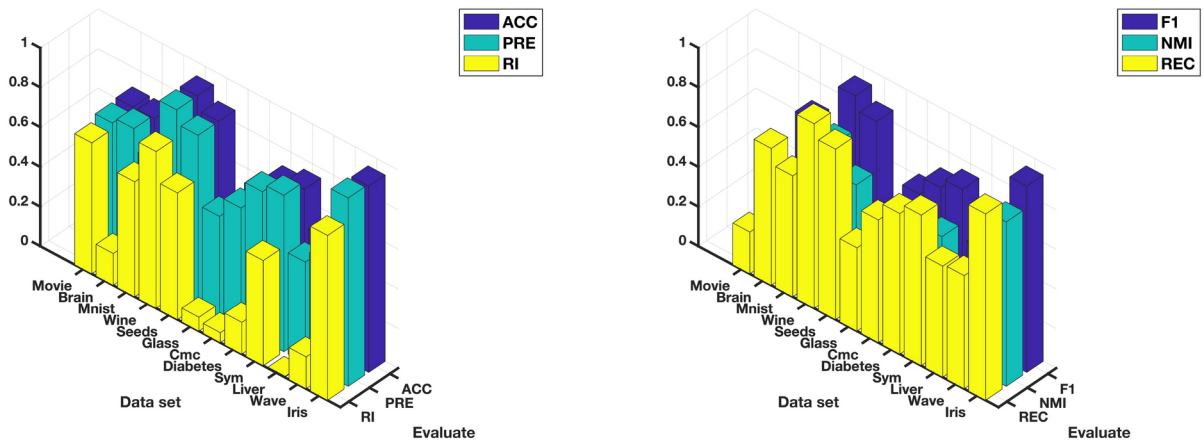

Fig. 3. Experimental results.

Datasets	Evaluate	KMEANS	FCM	DBSCAN	SPECTRAL	OPTICS	AUT-FCM	DMSC
	ACC	0.88	0.84	0.62	0.85	0.43	0.88	**0.94**
Iris	NMI	0.75	0.66	0.44	0.68	0.70	0.73	**0.83**
	PRE	0.90	0.84	0.43	0.87	0.66	0.89	**0.95**
	ACC	0.60	0.91	0.02	**0.91**	0.12	0.84	0.87
Seeds	NMI	**0.72**	0.71	0.04	0.71	0.26	0.57	0.63
	PRE	0.60	0.92	0.33	**0.91**	0.66	0.84	0.87
	ACC	0.66	0.67	0.00	0.64	0.63	0.68	**0.71**
Diabetes	NMI	0.02	0.08	0.00	0.00	0.01	0.07	**0.14**
	PRE	0.64	0.69	0.00	0.58	0.74	0.68	**0.75**
	ACC	0.54	0.58	0.01	**0.97**	0.36	0.80	0.58
Liver	NMI	0.02	0.04	0.01	**0.87**	0.72	0.27	0.03
	PRE	0.48	0.64	0.33	**0.98**	0.97	0.80	0.51
	ACC	0.62	0.55	0.71	0.70	0.13	0.62	**0.75**
Sym	NMI	0.39	0.65	**0.77**	0.68	0.38	0.46	0.59
	PRE	0.62	0.50	0.57	0.71	0.57	0.65	**0.76**
	ACC	0.50	0.61	0.10	0.49	0.01	0.62	**0.70**
Brain Tumor	NMI	0.00	0.06	0.09	0.01	0.12	0.02	**0.13**
	PRE	0.37	0.64	0.45	0.46	0.48	0.61	**0.72**
	ACC	0.41	0.34	0.35	0.40	0.15	0.26	**0.42**
Glass	NMI	0.17	0.19	0.27	0.24	0.18	0.25	**0.20**
	PRE	0.50	0.36	0.62	0.46	0.45	0.30	**0.51**
	ACC	0.70	0.95	0.33	0.93	0.39	0.81	**0.96**
Wine	NMI	0.42	0.84	0.12	0.80	0.28	0.54	**0.86**
	PRE	0.73	0.95	0.10	0.92	0.48	0.84	**0.96**
	ACC	0.61	0.66	0.00	0.50	0.50	0.61	**0.70**
Movie Reviews	NMI	0.04	0.08	0.00	0.00	0.00	0.03	**0.09**
	PRE	0.63	0.66	0.00	0.70	0.25	0.61	**0.69**
	ACC	0.53	0.55	0.00	0.58	0.53	0.59	**0.62**
Cmc	NMI	0.01	0.02	0.00	0.00	0.01	0.02	**0.03**
	PRE	0.54	0.59	0.00	0.56	0.59	0.59	**0.61**
	ACC	0.54	0.30	0.46	0.34	0.11	0.40	**0.63**
MNIST	NMI	0.51	0.22	0.29	0.48	0.00	0.34	**0.57**
	PRE	0.55	0.30	0.40	0.43	0.11	0.40	**0.63**
	ACC	0.50	0.33	0.30	0.49	0.33	0.49	**0.56**
Wave	NMI	0.19	0.36	0.03	**0.36**	0.00	0.31	0.17
	PRE	0.50	0.33	0.38	0.49	0.53	0.49	**0.57**

Table 3. Comparative experimental evaluation results. The bold font represents the best results in the comparative experiments.

range on most datasets is between 6% and 69%. Compared with autoencoder dimensionality reduction for clustering, the ACC improvement range is between 2% and 11%, and the PRE improvement range is between 2.0% and 26.7%. We speculate that this is because FCM only relies on fuzzy clustering and lacks this feature optimization mechanism, and relying solely on autoencoder dimensionality reduction in clustering methods may lose some important structural information, resulting in less stable and accurate clustering performance than our model. DMSC can closely integrate the clustering process with data reconstruction, continuously optimizing feature representations through multiple stages of clustering and data reconstruction to make them more suitable for clustering tasks. During the training process, our method focuses more on the distribution and structural model of the data. Through feature extraction and multiple clustering and data reconstruction during network training, the autoencoder can output features that are more conducive to clustering. This method not only reduces the dimensionality of the data, but also comprehensively captures the internal structure and distribution characteristics of the data, and feeds this information back into training and data reconstruction. Therefore, the extracted features are more discriminative, thereby improving the performance of the clustering model. For example, on the Iris dataset, the NMI of DMSC is 0.83, significantly higher than other methods. This may be because the feature distribution of this dataset is simple, and DMSC can more accurately capture inter cluster relationships through the synergistic effect of clustering and data reconstruction; On the Wave dataset, the ACC of DMSC is 0.56, which is lower but still better than other methods. This indicates that DMSC can still improve clustering performance through iterative optimization when the data feature distribution is uneven.

After the above experimental verification, our method has been validated on different types of datasets in multiple disciplines including Botany, Medicine, Chemistry, Sociology, and other fields. For example, this method can be effectively applied to plant clustering analysis in botany and medical image clustering processing in the field of life sciences, providing valuable reference and inspiration for related research. These application examples demonstrate that the proposed method has strong applicability and universality.

Convergence analysis

In this section, we independently verify the convergence efficiency of the Two-stage fuzzy C-means algorithm proposed in the deep multiple self-supervised clustering model for determining the initial cluster center and initial membership matrix, and compare it with the convergence efficiency of classical FCM.

In terms of time complexity, Two-stage FCM selects initial centers by maximizing the distance between data points. Since each time a new cluster center is selected during initialization, it is necessary to calculate the distance from all data points to the existing cluster centers, the time complexity of this step is $O(n \times c^2 \times d)$, where n is the number of data points, c is the number of clusters, and d is the data dimensionality. This step usually makes the cluster centers more uniformly distributed and closer to the true cluster centers, thereby significantly reducing the subsequent number of iterations T_{improved}. The time complexity of the classic FCM algorithm is $O(T_{\text{random}} \times n \times c \times d)$, where T_{random} is the number of iterations required for random initialization. Since the improved initial cluster centers are more optimal, T_{improved} is usually much smaller than T_{random}. Therefore, the total time complexity of the improved FCM algorithm is $O(n \times c^2 \times d + T_{\text{improved}} \times n \times c \times d)$. Further simplifying, in general, since T_{improved} is smaller, the time complexity of the Two-stage FCM can be approximated as $O(T_{\text{improved}} \times n \times c \times d)$. And because $T_{\text{improved}} < T_{\text{random}}$, the overall time complexity of the Two-stage FCM is lower compared to FCM, especially when the data scale is large.

Next, we will compare the convergence efficiency of the two methods through experiments. The experiment utilizes the Waveform dataset from the UCI database. Prior to analysis, data preprocessing was conducted, and for visualization purposes, the data was reduced to three dimensions using PCA. Both methods were configured with an equal number of clusters, an iteration convergence threshold of 0.05, and a maximum of 1000 iterations.

Figure 4 compares the clustering performance of the original FCM (fuzzy C-means) and the Two-stage FCM proposed in this paper at different iteration times. From the graph, it can be seen that as the number of iterations increases, the clustering results gradually tend to stabilize. In the clustering process of the original FCM, as the iteration progresses, the clustering centers gradually converge, but the clustering effect still changes significantly before higher iteration times, as shown in Fig. 4(a)–4(d), indicating that the algorithm needs more iterations to gradually improve the clustering effect of the data points. In the clustering process of Two-stage FCM, as shown in Fig. 4(e)–4(h), it can be seen that the clustering effect of data points is already quite obvious in the first and fifth iterations, indicating that Two-stage FCM can identify the internal structure of data well in the early stage, quickly adjust the allocation of data points, and achieve relatively stable clustering results with fewer iterations.

Through comparative experiments, we found that the traditional fuzzy C-Means clustering algorithm incurs higher computational costs and exhibits slower convergence of the objective function, stabilizing after 16 iterations. In contrast, the improved Two-stage fuzzy C-means clustering method proposed in this paper shows lower computational overhead and achieves stability after only 6 iterations. The comparison of accuracy between the two algorithms at the same number of iterations is shown in Table 4. In terms of iteration time,

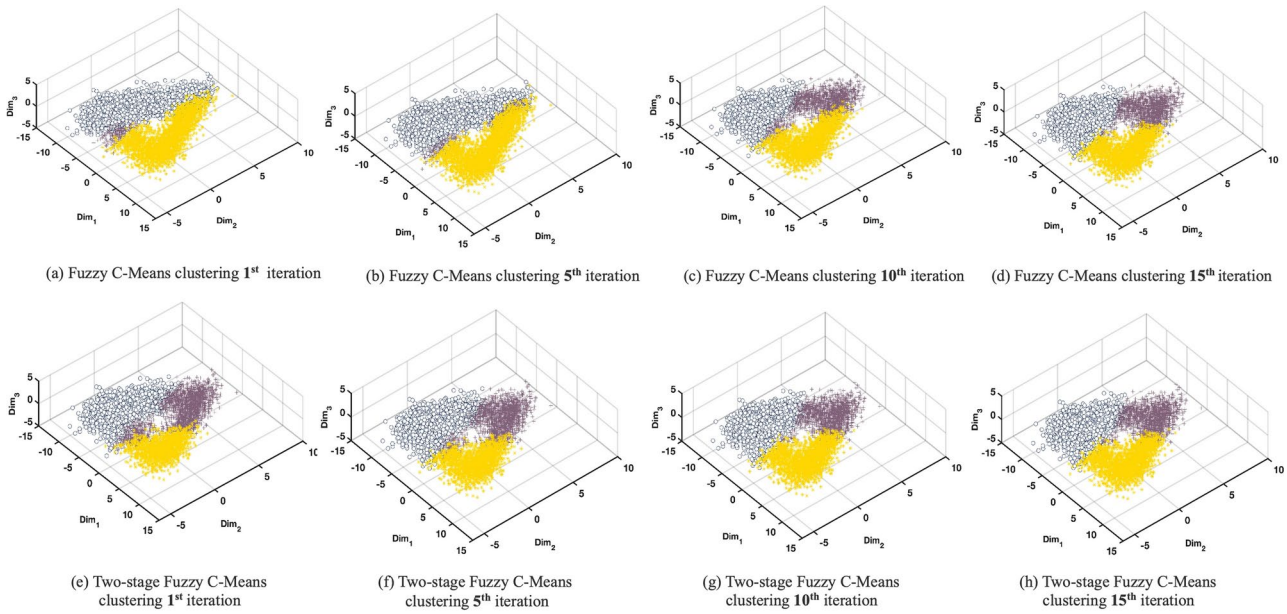

Fig. 4. Comparison of iterations between fuzzy C-means clustering and two-stage fuzzy C-means clustering.

Iteration times	1	3	5	7	10	15
Fuzzy C-Means	0.49	0.52	0.57	0.50	0.61	0.63
Two-stage fuzzy C-means(Two-stage FCM)	0.55	0.53	0.59	0.63	0.63	0.63

Table 4. Accuracy comparison under the same number of iterations.

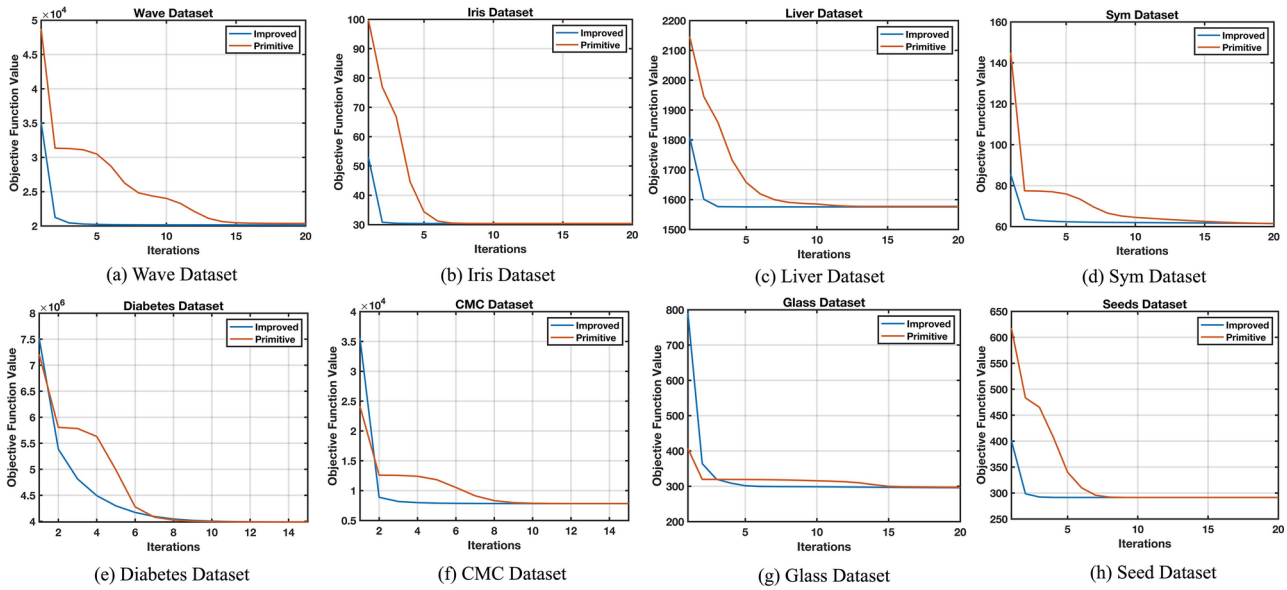

Fig. 5. Comparison of loss function iteration curves.

we conducted 20 experiments and averaged the time consumption. Compared with the original algorithm, the overall running time of Two-stage FCM has been reduced from 159.68ms to 125.37ms, which is 21.48% more efficient than traditional methods.

Additionally, We selected 8 UCI datasets for convergence analysis to ensure the representativeness and research efficiency of the results while simplifying complexity. These datasets are capable of covering different data features and application scenarios. The experimental results are shown in Fig. 5. For example, for the Iris dataset, the original fuzzy C-means clustering required 7 iterations for the objective function to gradually converge, while the improved method achieved stabilization after only 4 iterations. For the Liver dataset, the improvement was more significant, compared to the original method which required 13 iterations for convergence, the enhanced method completed clustering in just 5 iterations.

The experiments demonstrate that the Two-stage fuzzy C-means clustering method proposed in this paper performs well across most datasets, except for the Diabetes dataset where its performance was mediocre. This is speculated to be due to the imbalance in sample quantities among different classes in that dataset, which affects the algorithm's performance. On most datasets, the improved method shows significantly better efficiency in terms of iteration speed and convergence compared to the original algorithm.

Parameters analysis

Because the contour coefficient threshold ρ is a key parameter for data reconstruction in our proposed DMSC model, it determines when to perform clustering adjustments and reconstruction operations on the data. Therefore, in this section, we tested the robustness of the model to different contour coefficient thresholds ρ. Specifically, we observe the performance differences of the model under six different thresholds ρ by setting them.

Table 5 shows the experimental results of different contour coefficients on 8 datasets. It can be found that for most datasets, clustering performance significantly deteriorates when $\rho = 0.1$ or 0.3. The reason is that the contour coefficients at this time do not perform well. If data reconstruction is still carried out, it will inevitably be interfered by erroneous clustering, misleading the encoder to output incorrect feature representations, thereby reducing clustering performance. When $\rho = 0.9$, there is also a certain degree of performance decline, because the threshold requirement is too high, which limits the flexibility of training. However, when ρ is within a reasonable range, such as 0.5, 0.6, and 0.7, the performance of DMSC can maintain stability in experiments, which means that DMSC has a certain robustness in parameter selection.

Datasets	Evaluate	ρ values					
		$\rho = 0.1$	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.6$	$\rho = 0.7$	$\rho = 0.9$
Diabetes	ACC	0.49	0.50	0.71	0.70	0.68	0.70
	NMI	0.00	0.00	0.14	0.11	0.10	0.08
Sym	ACC	0.60	0.69	0.75	0.72	0.74	0.69
	NMI	0.44	0.39	0.59	0.42	0.55	0.38
Seeds	ACC	0.85	0.84	0.87	0.84	0.86	0.86
	NMI	0.62	0.62	0.63	0.58	0.61	0.62
Iris	ACC	0.78	0.92	0.94	0.96	0.91	0.87
	NMI	0.57	0.92	0.80	0.88	0.79	0.72
Wave	ACC	0.46	0.43	0.56	0.52	0.55	0.46
	NMI	0.13	0.07	0.16	0.13	0.36	0.15
Glass	ACC	0.34	0.38	0.42	0.42	0.46	0.37
	NMI	0.14	0.19	0.20	0.46	0.31	0.23
Liver	ACC	0.52	0.53	0.58	0.59	0.55	0.53
	NMI	0.00	0.00	0.03	0.04	0.00	0.00
Cmc	ACC	0.55	0.53	0.62	0.60	0.56	0.51
	NMI	0.00	0.00	0.03	0.02	0.01	0.00

Table 5. Performance comparison of multiple datasets under different ρ values.

Conclusion

We propose a Deep Multiple Self-Supervised Clustering (DMSC) model, which considers data distribution during the training of deep clustering networks. The model conducts multi-layer clustering evaluations throughout successive training rounds of the autoencoder network, employing a gradient-like approach for data reconstruction. This enables the autoencoder to learn features more conducive to clustering, thereby enhancing clustering performance. It is worth noting that our model exhibits significant superior performance compared to common and widely used clustering algorithms. Additionally, to enhance the efficiency of model training, we introduce a Two-stage clustering method within the model, leveraging initial cluster centers and membership matrices from fuzzy C-Means clustering. Comparative experiments reveal that this method significantly reduces the number of iterations and time per iteration in fuzzy C-Means clustering, facilitating faster convergence of the loss function and reducing computational overhead.

Although the proposed model has shown good performance on most datasets in experiments, its performance is not very excellent for some datasets. This study has not yet conducted a theoretical analysis of the performance differences between different datasets. In order to further enhance the adaptability and effectiveness of the model, future research plans will conduct detailed analysis on more diverse datasets and classify them based on their data distribution and intrinsic structure. Through this approach, we hope to identify which types of datasets can maximize the advantages of the proposed model and explore how to adjust the model's structure or optimization strategies for different types of datasets to improve their performance in various practical applications.

Data availability

The dataset used or analyzed in the current study can be downloaded for free from publicly accessible datasets https://github.com/liuliu-zufe/DMSC-experiment-data.

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Author contributions

L.Z. and G.L. finished conceptualization, methodology, supervision, project administration, resources offering and funding acquisition. G.L. , and Z.L. finished investigation. L.Z. , Z.L. and G.L. wrote the original draft preparation. L.Z. , and G.L. reviewed, edited and modification. Z.L. developed the software, validationformal analysis and visualization. L.Z. and Z.L. finished proofread. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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